Hiten Dalmia

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Education

Chennai Mathematical Institute — Chennai, India
Master of Science in Theoretical Computer Science (On Full Scholarship) — *Aug. 2026 – Discontinued*

Kirori Mal College, University of Delhi — Delhi, India
Bachelor of Science (Honours with Research) in Mathematics, Minor in Economics — *Nov. 2022 – May 2026*

Experience

Backend Engineer — Mar. 2023 – Present
Freelance & Fractal — *Remote*

- Delivered 10+ production web and backend applications for client businesses via Upwork and Fractal, specializing in system design, backend development, database architecture, and RESTful APIs.
- **Order Procurement & Management Solution** | *NodeJS, ExpressJS, PostgreSQL, Docker, RESTful APIs*
  - Designed backend RESTful APIs using ExpressJS, TypeScript, and PostgreSQL, containerizing services with Docker for reliable deployment on AWS cloud.
  - Built administrative order management endpoints supporting employee task dispatch, transactional audit logging, and recurring order schedules for client operations.
  - Constructed an automated order ingestion pipeline parsing unstructured incoming messages into PostgreSQL database records, eliminating manual data entry and cutting processing latency by 80%.
  - Structured relational PostgreSQL database schemas with B-tree indexing and foreign key constraints to maintain data integrity and optimize query execution speed.

Software Developer — Oct. 2023 – Apr. 2025
Tagglabs Experimental Pvt. Ltd. (Contract) — *Gurugram, Haryana (Remote)*

- **Cross-platform AR App** | *Django, C++, OpenCV, Flutter, RESTful APIs* **(Live link)**
  - Architected a native C/C++ bridge via `dart:ffi` to port OpenCV SIFT keypoint algorithms into Flutter, executing real-time object detection across Android and iOS.
  - Engineered a real-time camera feed processing pipeline normalizing OS frame buffers into C structs with zero memory leaks, maintaining 30+ FPS throughput.
  - Implemented real-time keypoint extraction, matching camera feed features against database records asynchronously to return spatial match coordinates.
  - Developed RESTful backend APIs in Django with PostgreSQL database queries to power location-aware spatial discovery features and user authentication for 1,000+ alpha users.
- **Kia National Conference App** | *Flutter, RESTful APIs, NodeJS*
  - Created backend API integrations and event management services for 500+ attendees at the Kia National Dealer Conference, handling real-time registration and attendee directories.

Technical Skills

Languages & Backend Systems: Python, JavaScript/TypeScript, C++, SQL, C, Dart, R, Mathematica
Frameworks & Databases: NodeJS, ExpressJS, PostgreSQL, Django, NestJS, React, NextJS, Flutter
Cloud, DevOps & Architecture: AWS, Docker, Linux, Bash, Git, RESTful APIs, System Design, CI/CD
Data Science & Tools: OpenCV, NumPy, Pandas, Scikit-learn, Vim, LaTeX, Agentic Development, n8n

Achievements

Erasmus Mundus Joint Master MATHS-DISC Fellowship — Europe
Issued by European Union and affiliated universities — *Mar. 2026*

Master in Mathematics (ISI) and MSc Computer Science (CMI) Scholarships — India
Indian Statistical Institute and Chennai Mathematical Institute respectively — *Jun. 2026*

Publications

Computable Analysis: Mathematical Foundations and Computability of Real Functions
Poincare Journal of Mathematical Physics & Applications, Vol. 1, No. 1 (2026), 1–20 (with Sujit Kumar)